



EGH404 Correlation and Regression

Faculty of Engineering

EGH404

Agenda

Correlation Analysis

Regression Analysis

Overview of the Research Process

Key Stages of the Research Process

- **Identifying the Problem**
Recognize a gap in knowledge or a specific issue that needs investigation.
- **Conducting a Literature Review**
Explore existing research to understand what has already been studied and where gaps exist.
- **Formulating Research Questions**
Develop clear, focused questions that guide the direction of the study.
- **Designing the Methodology**
Choose appropriate methods and tools to collect and analyze data.
- **Collecting Data**
Gather information through experiments, surveys, observations, or other techniques.
- **Analyzing Data**
Examine the data to identify patterns, relationships, and insights.
- **Drawing Conclusions**
Interpret findings to answer the research questions and understand implications.
- **Reporting Findings**
Share results through papers, presentations, or other formats to contribute to the field.



The research process (2021) courtesy of QUT Science and Engineering Faculty

Why Correlation & Regression Matter

Engineering research runs on data — sensor logs, lab measurements, field trials. Behind almost every study sits one recurring question:

How does one quantity depend on another?

- Detect relationships hidden in noisy measurements
- Quantify effects and predict outcomes you haven't tested
- Back design decisions with evidence, not intuition



CORRELATION

Is there a relationship?

Measures whether two variables move together, and how strong and in which direction.



REGRESSION

Quantify & predict it

Fits a model to estimate each factor's effect and forecast values you haven't observed.

What Correlation and Regression can Reveal?



Electrical

How does ambient temperature affect a battery's usable capacity?



Chemical

Which reactor conditions drive the highest product yield?



Civil / Structural

How does the water–cement ratio influence concrete strength?



Mechanical

How well does stress amplitude predict a part's fatigue life?



Environmental

How does river flow rate relate to downstream pollutant levels?

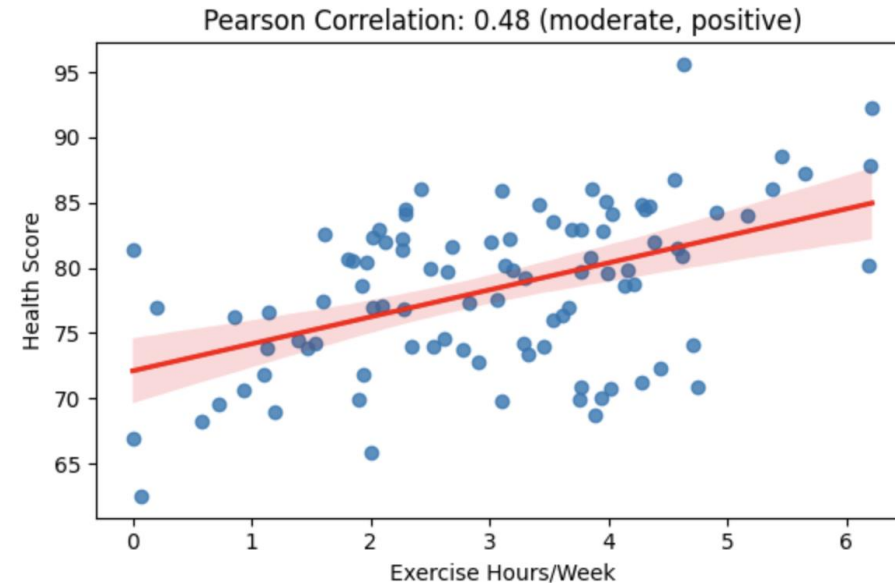


Materials

Which processing variables best predict a material's hardness?

Trend Line (in scatter plot)

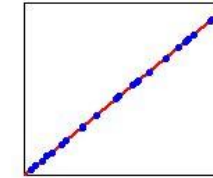
- **Purpose:** Visualize the general direction (or pattern) in the data between two variables.
- **What you can infer:**
 - **Positive slope:** As one variable increases, the other tends to increase (positive correlation).
 - **Negative slope:** As one variable increases, the other tends to decrease (negative correlation).
 - **Flat or noisy line:** Suggests weak or no linear relationship.



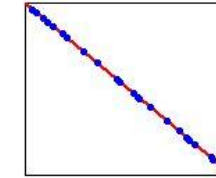
Pearson Correlation (r): Overview

- **Purpose:** Measures the *strength and direction* of a **linear relationship** between two continuous variables
- **Range:** -1 to $+1$
 - $r = +1$: Perfect positive linear relationship
 - $r = -1$: Perfect negative linear relationship
 - $r \approx 0$: No linear relationship
- **Interpretation:**
 - $|r|$ close to 1 $\rightarrow$ strong linear association
 - $|r|$ close to 0 $\rightarrow$ weak linear association
- **Key assumptions:**
 - Linear relationship
 - Continuous variables
 - No major outliers
- **Important note:**
 - Correlation **does not imply causation**

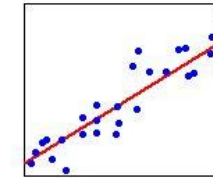
Pearson correlation = 1:



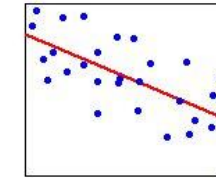
Pearson correlation = -1:



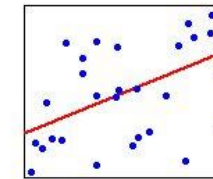
Pearson correlation = 0.9:



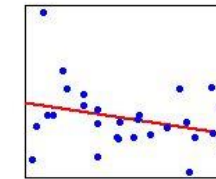
Pearson correlation = -0.7:



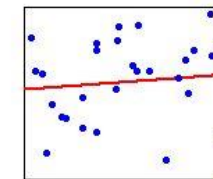
Pearson correlation = 0.5:



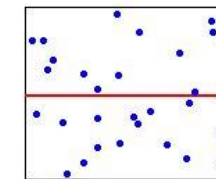
Pearson correlation = -0.3:



Pearson correlation = 0.1:



Pearson correlation = 0:



In Class Activity 1 – Pearson Correlation Analysis

LAB TOOL

Correlation Explorer

Pearson's r — When It Works, and When It Misleads

SCENARIOS

1 The Baseline

2 The Illusion

3 The Outlier Effect

4 The Subgroup Problem

5 The Nonsense Correlation

6 The Hidden Confounder

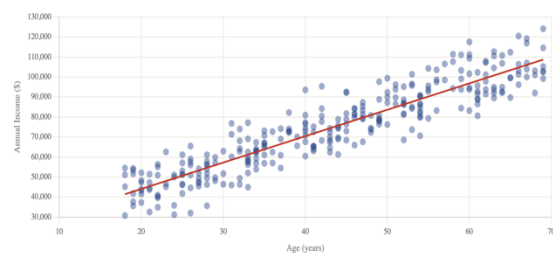
SCENARIO 1 OF 6 The Baseline

Age (years) × Annual Income (\$)

A survey of 300 working adults records their age and annual income. Before calculating anything, look at the scatter plot — what pattern do you see?

Scatter Plot with Trend Line

Age (years) vs Annual Income (\$)



Looking at the scatter plot above — what does it suggest?

By the end of this activity, students will be able to:

Interpret

Interpret the sign and magnitude of Pearson's r to describe the direction and strength of a linear relationship between two variables

Explain

Explain why a low r does not mean no relationship, and identify non-linear patterns that r cannot capture

Demonstrate

Demonstrate how a single extreme data point can substantially change r , and explain why this occurs mathematically

Recognise

Recognise when a statistically meaningful r may be misleading due to distinct subgroups within the data

Distinguish

Distinguish between a spurious correlation and a causal relationship, and identify plausible confounding variables that could explain an observed r

Evaluate

Evaluate whether Pearson's r is an appropriate summary statistic for a given dataset by inspecting the scatter plot before reporting r

Please use the [correlation_explorer.html](#) provided in practical's Canvas page for this activity

Single and Multiple Regression

Each component in the Regression summary table tells you different aspects about the model. So it's important to know what they tell you.

- **Coefficient:**
 - Represents the expected change in the response variable for a one-unit change in the predictor.
 - Tells us the *direction* (positive/negative) and *magnitude* of the relationship.
- **Standard Error:**
 - Measures the uncertainty in the estimated coefficient.
 - Smaller values indicate more precise and stable estimates.
- **p-value:**
 - Assesses whether the observed relationship is statistically meaningful.
 - A small p-value suggests the predictor is unlikely to have no effect.
- **F-statistic:**
 - Evaluates whether the model as a whole explains variability better than no model.
 - Useful for judging overall model usefulness, especially with multiple predictors.
- **Model Fit (R^2 , residuals):**
 - R^2 indicates how much of the variability in the data is explained by the model.
 - **Residuals** help diagnose whether model assumptions are reasonable or if patterns remain unexplained

```
In [8]: mod = smf.ols(formula='Lottery ~ Literacy + Wealth + Region', data=df)
```

```
In [9]: res = mod.fit()
```

```
In [10]: print(res.summary())
```

```
OLS Regression Results
=====
Dep. Variable:          Lottery    R-squared:                0.338
Model:                  OLS       Adj. R-squared:           0.287
Method:                 Least Squares   F-statistic:              6.636
Date:                  Sat, 28 Nov 2020   Prob (F-statistic):       1.07e-05
Time:                  14:39:43         Log-Likelihood:           -375.30
No. Observations:        85           AIC:                     764.6
Df Residuals:            78           BIC:                     781.7
Df Model:                6
Covariance Type:        nonrobust
=====
                    coef    std err          t      P>|t|      [0.025     0.975]
-----
Intercept          38.6517     9.456      4.087     0.000     19.826     57.478
Region[T.E]       -15.4278     9.727     -1.586     0.117    -34.793     3.938
Region[T.N]       -10.0170     9.260     -1.082     0.283    -28.453     8.419
Region[T.S]        -4.5483     7.279     -0.625     0.534    -19.039     9.943
Region[T.W]       -10.0913     7.196     -1.402     0.165    -24.418     4.235
Literacy           -0.1858     0.210     -0.886     0.378     -0.603     0.232
Wealth             0.4515     0.103      4.390     0.000      0.247     0.656
=====
Omnibus:                 3.049   Durbin-Watson:           1.785
Prob(Omnibus):           0.218   Jarque-Bera (JB):         2.694
Skew:                    -0.340   Prob(JB):                 0.260
Kurtosis:                 2.454   Cond. No.                 371.
=====
```

In Class Activity 2 – Regression Analysis

By the end of this activity, students will be able to:

Interpret

Interpret a regression coefficient in plain English, including its direction, magnitude, and units

Explain

Explain what standard error indicates about the reliability of a coefficient estimate

Use

Use the p-value to assess whether a predictor contributes meaningfully to a model

Interpret

Interpret R^2 as the proportion of variance explained, and identify what a low R^2 implies

Identify

Identify a spurious or redundant predictor using p-values and changes in Adjusted R^2

Distinguish

Distinguish between a well-specified model and one that is underfitting or overspecified

LAB TOOL Regression Explorer *Linear Regression — Single and Multiple Predictors*

1 UNDERSTANDING THE STATISTICS 2 BUILDING A MULTIPLE REGRESSION MODEL 3 BUILD YOUR BEST MODEL

STATISTICS

- 1 Coefficient
- 2 Standard Error
- 3 p-value
- 4 R^2
- 5 Residuals
- 6 F-statistic

This panel shows a single regression model predicting **Income** from **Age**. The table below displays five key statistics from the regression output. Each row corresponds to one concept — click an unlocked row to open an in-depth panel exploring what that statistic means and how to interpret it. Complete each panel in order to unlock the next.

Regression Output — Income - Age		
Statistic	Value	Plain-English Definition
Coefficient (β_1)	+1,314	Expected change in outcome per one-unit increase in the predictor. Direction (+) and magnitude.
Standard Error	29	Uncertainty in the coefficient estimate. Smaller = more precise and stable.
p-value	< 0.001	Probability of observing this coefficient by chance if there were no real relationship.
R^2	0.873	Proportion of variability in the outcome explained by the model (0 to 1).
Residuals	RMSE: 7,669	Actual minus predicted values. Patterns in residuals reveal model problems.
F-statistic	2,052	Tests whether the model as a whole explains significantly more variance than no model.

↑ Click the Coefficient row to open the first panel.

Dataset: 300 simulated survey respondents, age 18–70. Income modelled as $\text{age} \times 1,200 + \text{age}^{1.5} \times 10 + \text{noise}$ ($r = \$8,000$).

PROGRESS — 0 / 6

Please use the [regression_explorer.html](#) provided in practical's Canvas page for this activity

Use the Interactive Quiz Agent to Test Your Knowledge

Test Your Knowledge

Ready to Test Your Knowledge? Try This Quiz Agent!



Hi! I'm your adaptive data analysis quiz tutor. Would you like to try a few questions to test your understanding? [Hide message](#)



yes



I am confident in responding to your query using the information provided in the **EXTRA_CONTEXT** section from the **RESOURCES** tag. Please let me know how I can assist you!



I need questions in correlation



Which topic would you like to practise? (Processing Data, Visualisation, Correlation, Regression)



Correlation



Thank you for selecting the topic "Correlation." I will now retrieve a random question from the correlation file.

Question:

.....

Enter your message



You can access this agent through this week's practical page in Canvas:
https://canvas.qut.edu.au/courses/24207/pages/8-dot-3-practical-correlation-and-regression?module_item_id=2111748

Wrap-Up

Key Takeaways

- **Correlation vs. Causation:**
 - Correlation shows association, not cause-and-effect. Always interpret carefully.
- **Interpreting Coefficients:**
 - Coefficients indicate the direction and magnitude of relationships in regression models.
 - Check p-values and confidence intervals for significance.
- **Regression:**
 - Regression helps predict outcomes and quantify relationships between variables.
 - Understand assumptions (linearity, independence, homoscedasticity) for valid models.

Homework

- **Work on the correlation and regression tasks in Assignment 2:**